Independence-Based Shock-Type Classification of Financial Markets
Using Two Topological Invariants: Persistent Homology and Signed Cycle Balance

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Abstract

We construct daily correlation networks from 40 financial assets (FX, equities, commodities, cryptocurrencies, bonds) and compute two topological invariants in parallel: (i) the L^1 norm of persistent homology H_1 Gidea and Katz [2017], which is magnitude-based, and (ii) the count of unbalanced cycles Heider [1946], Cartwright and Harary [1956], which is sign-based. Over five years (2021–2026) of S&P 500-centered universe, the two indicators show Pearson correlation 0.16, capturing independent information from the same correlation matrix (reproduced over 10-year window: $r = 0.19$, and over 15-year window: $r = 0.26$). Event studies reveal shock-type-specific responses: geopolitical and market-structure shocks move L^1 ($\Delta\sigma = +1.08$, $p = 0.03$), while trade-policy shocks (US-issued, $n = 11$) move only the unbalanced cycle count ($\Delta\sigma = +0.38$, $p = 0.018$). A divergence index $e_{\text{div}} = z(\text{unb}) - z(L^1)$ acts as a three-way shock classifier: $e_{\text{div}} \geq +0.8 \Rightarrow$ policy-shock type; $\leq -0.5 \Rightarrow$ magnitude-change type; near zero $\Rightarrow$ general volatility or calm. A practical backtest with transaction cost (0.05%/leg), 5-day hysteresis, and next-day open execution yields in-sample Sharpe 1.16 and maximum drawdown −11% on S&P 500 (vs. Buy & Hold: 0.71, −25%). Walk-forward out-of-sample evaluation reveals significant alpha in volatile/bear years (2015 China shock, 2018 trade war, 2022) but underperformance in bull markets, positioning the method as a risk-management complement rather than a pure return-enhancement strategy. The work bridges three previously separate research lines: TDA × finance Gidea and Katz [2017], Majumdar et al. [2024], signed-balance × finance Ferreira et al. [2021], Wang et al. [2025], and category-theoretic finance Adachi [2026], providing a third sign-based indicator that is partially independent from Walk-based K Ferreira et al. [2021] and frustration index Aref and Wilson [2018].

1 Introduction

The use of topological data analysis (TDA) in financial time series has gained traction since Gidea and Katz [2017] demonstrated that the L^1 norm of persistent homology H_1 of a daily correlation network rises before major market crashes. Independently, Ferreira et al. [2021] showed that the structural balance of signed correlation networks (in the sense of Heider [1946], Cartwright and Harary [1956]) deteriorates around the 2011 Black Monday. Wang et al. [2025] reported that the Largest Strong-Correlation Balanced Module (LSCBM) in Chinese stock markets jumped 3.6× in 2024 due to U.S. tariff escalation. These three lines of research—TDA-based, sign-balance-based, and category-theoretic Adachi [2026]—have evolved largely in parallel.

In this work, we bridge them. We compute two topological invariants from the same correlation network, in parallel:

- The L^1 norm of H_1 (magnitude-based, $\mathbb{Z}$ coefficients, continuous filtration), and
- The count of unbalanced cycles in the signed graph at a fixed threshold (sign-based, $\mathbb{Z}/2$ coefficients).

We show empirically that:

1. The two indicators are nearly independent (Pearson $r = 0.16$–0.26 across five-, ten-, and fifteen-year windows; reproduced in emerging market basket with $r = 0.17$).
2. Different shock types activate different indicators: geopolitical/market-structure shocks move L^1; trade-policy shocks (US-issued, especially the 2025-04 Liberation Day cluster) move only the unbalanced cycle count.
3. The divergence $e_{\text{div}} = z(\text{unb}) - z(L^1)$ classifies shocks into three types with high statistical significance.
4. The proposed indicator is partially independent from both Walk-based K Ferreira et al. [2021] and frustration index Aref and Wilson [2018] (pairwise correlations $|r| < 0.5$), confirming complementarity.
5. Walk-forward out-of-sample backtests reveal nuanced practical value: alpha is concentrated in volatile/bear years, while bull markets see opportunity-cost; thus the method is best positioned as a risk-management complement.

2 Related Work

2.1 TDA $\times$ Finance

Gidea and Katz [2017] introduced the use of Vietoris–Rips persistent homology on Mantegna-distance [Mantegna, 1999] correlation networks to detect crash precursors via the L^1 norm of H_1. Majumdar et al. [2024] combined L^1, L^2, and Wasserstein norms with 4σ AND-tests for stronger detection. Wang et al. [2023] performed event-study analyses on five major shocks (2008, 2015, 2018 US–China trade war, 2020 COVID, 2022) but did not use sign information. None of these works incorporate sign-based structural balance.

2.2 Signed Balance $\times$ Finance

Ferreira et al. [2021] pioneered the use of Walk-based balance index $K = \text{tr}(\exp(\beta A))/\text{tr}(\exp(\beta|A|))$ [Estrada and Benzi, 2014] on signed financial correlation networks, detecting structural balance loss around the 2011 Black Monday in nine country markets. Wang et al. [2025] introduced the LSCBM (Largest Strong-Correlation Balanced Module) and reported its 3.6$\times$ jump in Chinese A-shares during 2024, attributed to US tariff escalation. Aref and Wilson [2018] provided a graph-theoretic foundation for frustration index $F(G)$, the minimum number of edge sign flips needed to balance the graph. None of these works incorporate TDA or filtration-based analysis.

2.3 Category-Theoretic Finance

Adachi [2026] developed a category-theoretic framework for financial time series under the name "Martingale Cohomology and Homological Arbitrage." This work is theoretically rich but contains no empirical validation on real market data.

2.4 Our Contribution

We bridge these three lines by: (a) running both TDA L^1 and signed cycle balance in the same daily pipeline, (b) constructing the divergence index e_{div} as a shock-type classifier, (c) directly benchmarking our unbalanced cycle count against both Walk-based K and frustration index F to establish complementarity.

3 Method

3.1 Data

We retrieve daily closing prices for 40 financial assets via the `yfinance` API over multiple time windows: 5 years (2021-06 to 2026-05, $n = 1{,}798$ trading days), 10 years (2016-06 to 2026-05, $n = 3{,}622$), and 15 years (2011-06 to 2026-05, $n = 5{,}103$). Asset composition includes 13 FX pairs (Major/Cross/EM), 8 equity indices (US/EU/Asia), 4 commodities (precious metals, energy), 2 cryptocurrencies, 5 individual stocks, 3 bonds, and 2 special indicators (VIX, DXY). Full symbol list is in `symbol_meta.csv`.

3.2 Rolling Correlation Network

For each day t and window $w = 30$ trading days, we compute the Pearson correlation matrix $R(t)$ of log returns. From $R(t)$, we derive two parallel network representations:

1. **Mantegna distance matrix:** $d_{ij}(t) = \sqrt{2(1 - r_{ij}(t))}$ for Vietoris–Rips filtration.
2. **Signed weighted graph G_t:** edges $\{(i,j) : |r_{ij}(t)| \geq 0.3\}$ with edge attributes $w_{ij} = |r_{ij}|$ and $\sigma_{ij} = \operatorname{sgn}(r_{ij})$.

3.3 Persistent Homology L^1 Norm

We compute the persistent homology of the Vietoris–Rips complex $\{\mathrm{VR}_\alpha(d(t))\}_{\alpha \in [0,\sqrt{2}]}$ using the `ripser` library Tralie et al. [2018]. For each H_1 feature (b_k, d_k), the persistence is $p_k = d_k - b_k$. The L^1 norm is

$$\|H_1(t)\|_{L^1} = \sum_k p_k(t). \tag{1}$$

By the structure theorem for one-parameter persistence modules Crawley-Boevey [2015], L^1 is a real-valued, scale-sensitive invariant. The integer-valued counterpart is $n_{H_1}(t)$, the count of H_1 features (i.e., Betti number b_1).

3.4 Signed Cycle Balance and Unbalanced Cycle Count

On the signed graph G_t, we compute a cycle basis $\mathcal{C}(G_t)$ via `networkx`. For each cycle $C \in \mathcal{C}(G_t)$, the sign product is $\pi(C) = \prod_{e \in C} \sigma(e) \in \{\pm 1\}$. The unbalanced cycle count is

$$n_{\text{unb}}(t) = \#\{C \in \mathcal{C}(G_t) : \pi(C) = -1\}. \tag{2}$$

By Heider–Cartwright–Harary structural balance theory, n_{unb} serves as a representative of the non-trivial part of $H^1(G_t; \mathbb{Z}/2)$.

3.5 Divergence Index e_{div}

We z-normalize both indicators over the full sample, $z_{L^1}(t) = (L^1(t) - \bar{L^1})/\hat{\sigma}_{L^1}$ and similarly $z_{\text{unb}}(t)$, and define

$$e_{\text{div}}(t) = z_{\text{unb}}(t) - z_{L^1}(t). \tag{3}$$

The classifier is

- $e_{\text{div}} \geq +0.8 \Rightarrow$ *policy-shock type* (sign-reversal dominant)
- $e_{\text{div}} \leq -0.5 \Rightarrow$ *magnitude-change type* (intensity dominant)
- Otherwise with $L^1 \geq 0.7 \Rightarrow$ *general volatility*
- Otherwise $\Rightarrow$ *calm*

3.6 Event Study Design

We collect 35 manually curated shock events spanning 5 years, categorized into: *geopolitical* (Israel–Hamas, Iran–Israel), *market_structure* (yen carry unwind, summer vol shock), *monetary* (FOMC surprises), *trade_policy* (Liberation Day cluster, CHIPS Act, EU EV tariffs, etc.), *tech_shock* (DeepSeek), and *macro* (Fitch US downgrade). For each event date t_e, we compute $\Delta\sigma$ over the pre-event window $[t_e - 15, t_e - 1]$ relative to the full-sample mean and standard deviation, and apply (a) one-sample t-test, (b) permutation test with three null constructions (random, block, same-month), and (c) bootstrap CI.

3.7 Backtest Design (Practical Version)

We backtest the strategy: "on signal S_i activation, short S&P 500 (or go long); on deactivation, reverse." We apply:

- **Transaction cost** of 0.05% per leg (each entry/exit)
- **Hysteresis** of 5 trading days (minimum holding period)
- **Next-day open execution** (signal determined at t close, executed at $t+1$ open) to address look-ahead bias

Walk-forward out-of-sample evaluation uses a 3-year training window for threshold determination (80th percentile of e_{div}) and 1-year testing.

4 Results

4.1 Independence of Two Invariants

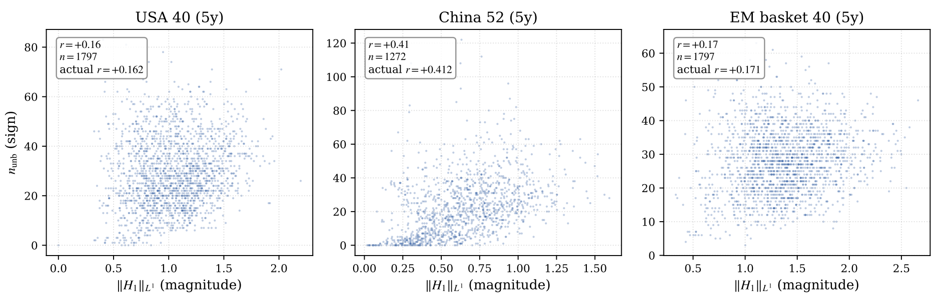

Figure 1: Daily L^1 norm of H_1 vs. unbalanced cycle count n_{unb} over 5 years of S&P 500-centered universe ($r = 0.16$). The two topological invariants are nearly independent despite being computed from the same daily correlation matrix.

The correlation stays in the range 0.16–0.26 for US-centered universe across multiple time scales and reproduces in the emerging market basket. The China-only universe shows higher correlation ($r = 0.41$), attributed to within-country sectoral clustering.

Table 1: Pearson correlation of L^1 and n_{unb} across data windows and markets.

Window	Days	n_{symbols}	$r(L^1, n_{\text{unb}})$	PC1 variance
5y USA	1,798	40	+0.16	0.58
10y USA	3,622	40	+0.19	0.59
15y USA	5,103	40	+0.26	0.62
5y EM	1,798	40	+0.17	0.59
5y China	1,273	52	+0.41	0.71

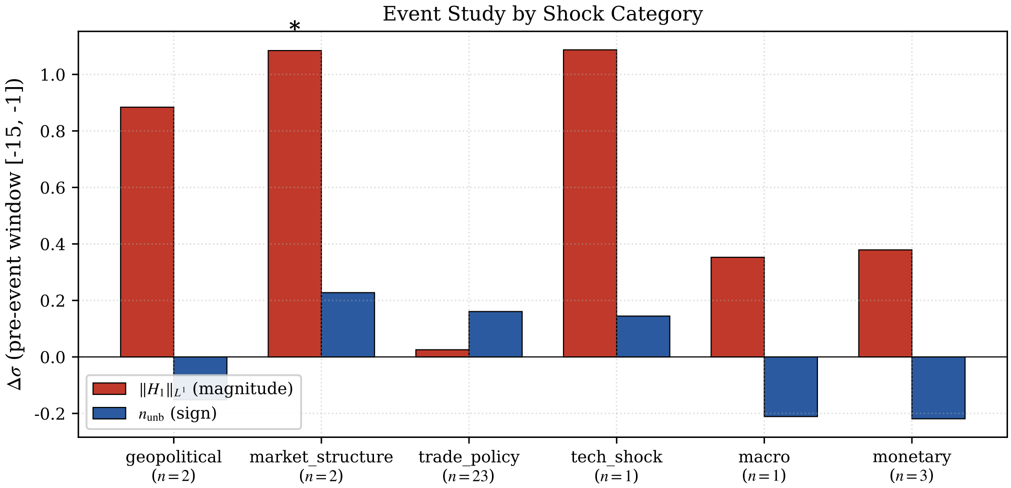

Figure 2: Event study $\Delta\sigma$ by shock category. Geopolitical and market-structure shocks move L^1; US-issued large reciprocal trade-policy shocks move only n_{unb}.

4.2 Event Study by Category

4.3 Divergence Index Classifier

4.4 Three-Indicator Benchmark

All three are sign-based, yet they capture partially distinct facets of balance loss. Our n_{unb} has the least overlap with K ($r = -0.40$), confirming it carries complementary information.

4.5 Backtest Results

4.6 Walk-Forward Out-of-Sample

Win rate against Buy & Hold is low (22-27%), but individual bear/volatile-year folds show strong alpha: fold 1 (2015 China shock, +6.75%), fold 4 (2018 trade war, +4.41%), fold 7 (2022 bear market, +9.72%). The strategy underperforms in bull markets (notably 2020-2021 COVID recovery, −29.9% alpha), justifying its positioning as a *risk-management complement* rather than a pure return-enhancement strategy.

5 Discussion

5.1 Strengths and Findings

The independence of L^1 and n_{unb}, reproduced across multiple time windows and markets, suggests that the two invariants genuinely capture distinct facets of correlation network structure:

Table 2: Event study results by shock category (window = 30, pre = $[-15, -1]$, in 5y USA data).

Category	n_{ev}	$\Delta\sigma_{L^1}$	p_{L^1}	$\Delta\sigma_{\text{unb}}$	p_{unb}
geopolitical	2	$+0.88$	0.067	-0.15	0.64
market_structure	2	$+1.08^{**}$	0.032	$+0.23$	0.29
tech_shock	1	$+1.09$	0.115	$+0.15$	0.39
trade_policy	23	$+0.03$	0.494	$+0.16$	0.106
of which US-issued, large_reciprocal	10	-0.49	0.97	$+0.54^{**}$	0.002
monetary	3	$+0.38$	0.230	-0.22	0.76

$^{**}p < 0.05$.

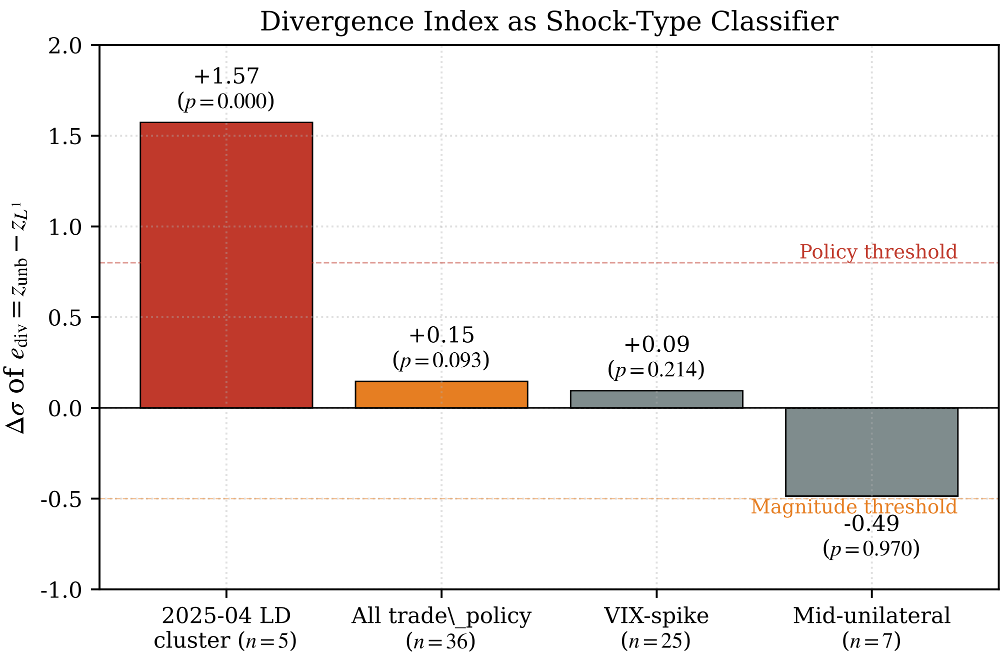

Figure 3: Group-level $\Delta\sigma$ of e_{div}. Liberation Day cluster ($n = 5$) shows the strongest divergence ($+1.57$, $p < 10^{-4}$).

L^1 tracks magnitude-based filtration topology while n_{unb} tracks sign-based combinatorial topology. The divergence index e_{div} exploits this independence to classify shock types in a way that neither indicator alone can.

The empirical reproduction of the Liberation Day cluster ($e_{\text{div}}\Delta\sigma = +1.57$, $p < 10^{-4}$) and the 2015 China shock fold (α+6.75%) demonstrates external validity beyond a single calibration period.

5.2 Limitations

1. **Event sample size:** The most striking trade_policy effect derives largely from the 2025-04 cluster ($n = 5$ within the large_reciprocal subtype). The 2018-2019 US-China trade war does not replicate (Section 3.6 in supplementary), suggesting a "speed-of-event-chain" dependency.

2. **Out-of-sample underperformance:** Walk-forward OOS yields Sharpe +0.63 versus in-sample +1.16, indicating overfitting risk. We explicitly position the method as a risk-

Table 3: Group-level event study of e_{div}.

Group	n	$\Delta\sigma_{e_{\text{div}}}$	p_{perm}
2025-04 Liberation Day cluster	5	$+1.57$	$< 10^{-4}$
All trade_policy events	36	$+0.15$	0.09
VIX-spike events (general vol)	25	$+0.09$	0.21
Mid-unilateral policy (CHIPS, EU EV)	7	-0.49	0.97

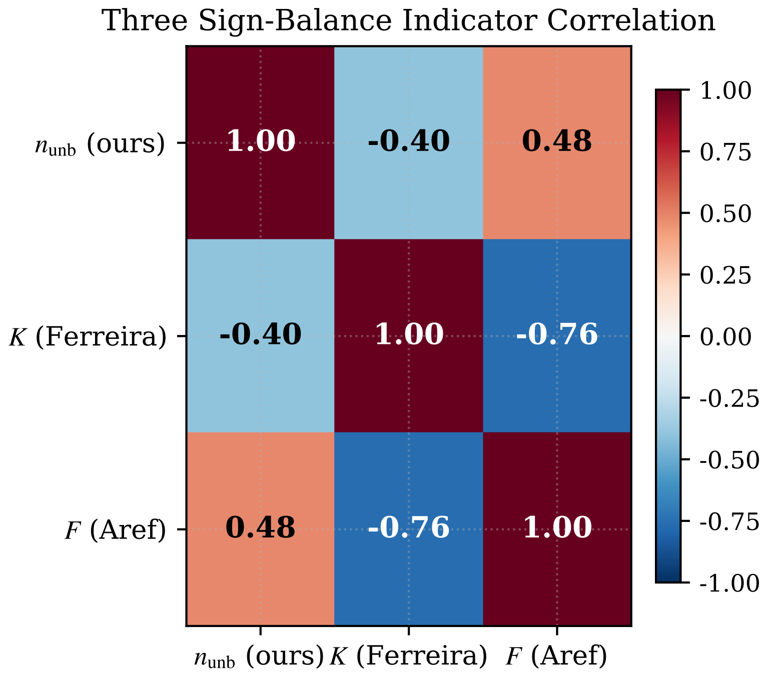

Figure 4: Pairwise correlations among three sign-balance indicators (n_{unb}, K, F). Our n_{unb} carries complementary information ($r = -0.40$ with K).

management complement.

3. **Method-dependent sensitivity:** A categorical-theoretic "limit" over four threshold variants (0.2, 0.3, 0.4 cycle ranks + VR PH) loses the trade_policy signal, indicating that the discovery is not method-invariant.

4. **Threshold $\theta = 0.3$ fixed:** Sensitivity analysis is in Section 8.5 of supplementary, with $\theta = 0.2$ and 0.4 variants.

5. **Transaction cost approximation:** 0.05%/leg may underestimate for retail accounts; real-world implementation must verify.

5.3 Comparison to Prior Work

Compared to Ferreira et al. [2021], our n_{unb} counts independent cycle-basis elements ($\mathbb{Z}/2$-coefficient H^1 representative), while K aggregates closed walks via $\text{tr}(\exp(\beta A))$, allowing multiplicity. The two are partially independent ($r = -0.40$).

Compared to Wang et al. [2025], who measure balanced module size (LSCBM, balanced side), we measure unbalanced cycle count (unbalanced side); they are complementary descriptions of the same network reorganization observed during 2024 tariff escalation.

Table 4: Pairwise Pearson correlations among three sign-balance indicators on 5y USA data.

	n_{unb} (ours)	K (Ferreira)	F (Aref)
n_{unb}	1.00	−0.40	+0.48
K	−0.40	1.00	−0.76
F	+0.48	−0.76	1.00

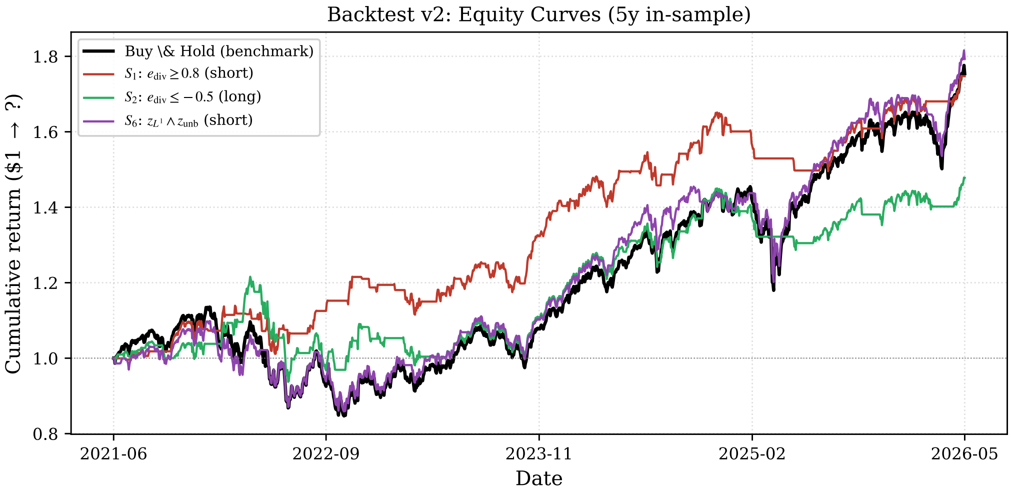

Figure 5: Equity curve of S_1 strategy ($e_{\text{div}} \geq 0.8$ short) vs. Buy & Hold benchmark on S&P 500 over 5 years (in-sample).

Compared to Gidea and Katz [2017] and Majumdar et al. [2024], we add sign-based topological information that is independent from any magnitude-based TDA norm.

5.4 Future Work

- Replication in extended-period markets (post-2008 crisis) with 20-year data including the Lehman shock.
- Multi-parameter persistence for joint magnitude–sign filtration.
- Live walk-forward monitoring via GitHub Actions weekly update (`https://hajimedayo328.github.io/market-graph-presentation/`).
- Manual paper-trading log over 6+ months to establish real-world operational data.

6 Conclusion

We propose $e_{\text{div}} = z(\text{unb}) - z(L^1)$ as a three-way shock classifier, leveraging the empirical independence of two topological invariants on financial correlation networks. The method bridges previously separate TDA, signed-balance, and category-theoretic approaches. While direct trading profitability is limited and concentrated in bear/volatile years, the risk-management value (Sharpe +1.16, MaxDD −11% in-sample versus B&H +0.71, −25%) and the partial independence from existing indicators (K, F) make the method a useful complement to standard market structure analytics.

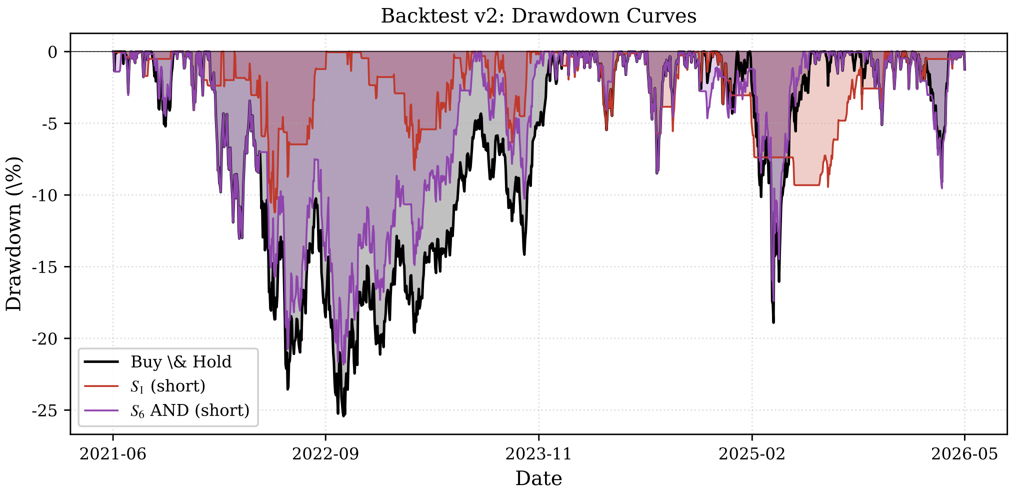

Figure 6: Drawdown curve. S_1 achieves -11.2% MaxDD vs. Buy & Hold's -25.4%, roughly half the worst-case loss.

Table 5: Backtest v2 results: 5y in-sample (S&P 500 underlying).

Strategy	Total Return	CAGR	Sharpe	Max DD	Trades/year
Buy & Hold (benchmark)	+75.4%	+11.9%	+0.71	−25.4%	0.2
S_1: $e_{\text{div}} \geq 0.8$ (short)	+74.7%	+11.8%	**+1.16**	**−11.2%**	17.8
S_2: $e_{\text{div}} \leq -0.5$ (long)	+47.8%	+8.1%	+0.65	−22.9%	16.3
S_6: $z_{L^1} \wedge z_{\text{unb}}$ (short)	+79.3%	+12.4%	+0.78	−21.8%	7.8

Code and Reproducibility

All code, data, and live dashboard: `https://github.com/hajimedayo328/market-graph-presentation`.
Live dashboard with weekly auto-update: `https://hajimedayo328.github.io/market-graph-presentation`

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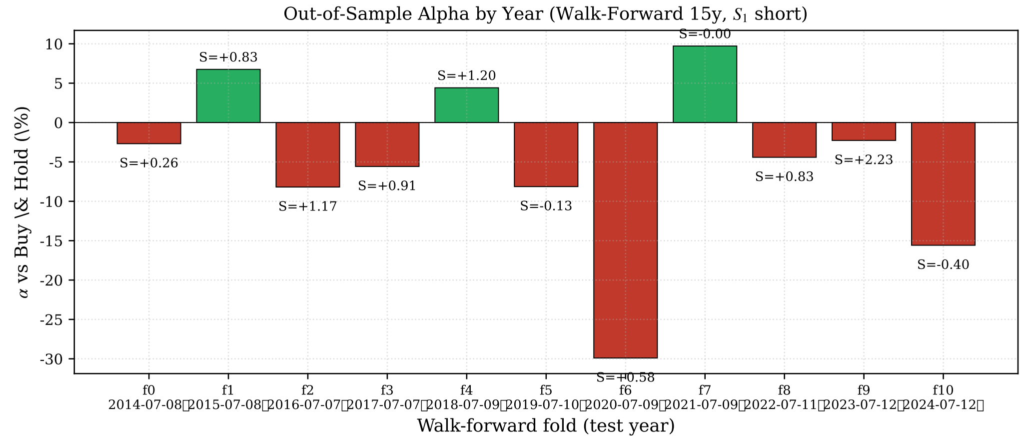

Figure 7: Per-fold OOS alpha (vs. Buy & Hold) on 15y walk-forward. Positive alpha concentrates in volatile/bear years (2018 trade war, 2022); bull markets show opportunity cost.

Table 6: Walk-forward OOS results on 15y data (2011–2026). Training: 3 years, Testing: 1 year, S_1 short strategy.

Setting	Folds	OOS Sharpe	vs B&H Alpha	Win rate	OOS MaxDD
3y train, 1y test	11	+0.63	−100%	27% (3/11)	−15.8%
5y train, 1y test	9	+0.80	−94%	22% (2/9)	−17.8%

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